

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2455634.949 +/- 0.075 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.082 +/- 0.052
Transit depth (Rp/Rs)^2	0.68 +/- 0.86 [%]
Orbital Inclination (inc)	88.43 +/- 2.11
Airmass coefficient 1 (a1)	1.0321 +/- 0.0084
Airmass coefficient 2 (a2)	-0.021 +/- 0.029
Scatter in the residuals of the lightcurve fit is	0.81 %
Best Comparison Star	None
Optimal Aperture	4.11
Optimal Annulus	11.39
Transit Duration (day)	0.059 +/- 0.038

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.082 $\pm$ 0.052 vs 0.1457 (deviation 0.96 σ)
Duration fit vs published	0.059 d vs 0.12 d (deviation 1.25 σ)
Mid-time O-C	-1.1 min
Depth SNR	1.2
Residual scatter	0.81 %

Light curve

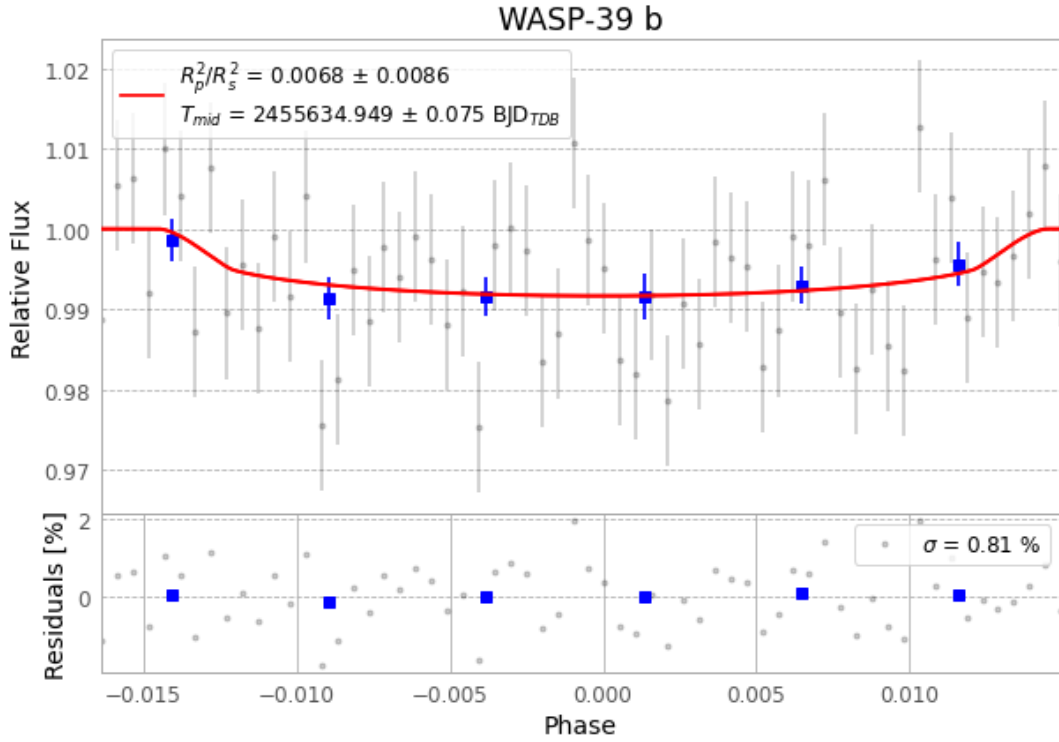


Figure 1 — FinalLightCurve_WASP-39 b_2011-03-14.png (SHA-256 fa5d282c6e522db8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [146, 151], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6311c0ffe201f01d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

62 FITS frames (41 MB), WASP-39110314090312.FITS → WASP-39110314120613.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-110314.log (e18dd60e450c...), FinalLightCurve_WASP-39 b_2011-03-14.png (fa5d282c6e52...), FinalLightCurve_WASP-39 b_2011-03-14.csv (6827e57caa2b...)

Compute resources

Wall time 97.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-13 00:29Z.